RIEFFEL'S THEOREM ON THE BICOMMUTANT OF A RIGHT IDEAL

THE STACKS PROJECT AUTHORS

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Editorial note. The proof is the author's complete argument; the author even pauses in a footnote to fix the convention for the right module structure over the bicommutant, and every step of the argument is written out in the counted span.

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Allowed standard prerequisites (editorial).

- Endomorphism rings and bicommutants. Same chapter, the definitions of the endomorphism ring of a module and of the bicommutant, and the convention for the module structure over them. Supplies the objects A' and A" of the proof and the side on which the bicommutant acts.
- Two-sided ideals and the product ideal. Same chapter, the elementary theory of two-sided ideals in a ring. Supplies that the product of the ring with a nonzero right ideal is a two-sided ideal, and that a ring with no nontrivial two-sided ideal has an injective structural map determined by the identity.

Lemma 0.1. *Let A be a possibly noncommutative ring with 1 which contains no nontrivial two-sided ideal. Let M be a nonzero right ideal in A, and view M as a right A-module. Then A coincides with the bicommutant of M.*

Proof. Let $A' = \mathrm{End}_A(M)$, so M is a left A'-module. Set $A'' = \mathrm{End}_{A'}(M)$ (the bicommutant of M). We view M as a right A''-module[1]. Let $R : A \to A''$ be the natural homomorphism such that $mR(a) = ma$. Then R is injective, since $R(1) = \mathrm{id}_M$ and A contains no nontrivial two-sided ideal. We claim that $R(M)$ is

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[1]This means that given $a'' \in A''$ and $m \in M$ we have a product $ma'' \in M$. In particular, the multiplication in A'' is the opposite of what you'd get if you wrote elements of A'' as endomorphisms acting on the left.

a right ideal in A''. Namely, $R(m)a'' = R(ma'')$ for $a'' \in A''$ and m in M, because *left* multiplication of M by any element n of M represents an element of A', and so $(nm)a'' = n(ma'')$ for all n in M. Finally, the product ideal AM is a two-sided ideal, and so $A = AM$. Thus $R(A) = R(A)R(M)$, so that $R(A)$ is a right ideal in A''. But $R(A)$ contains the identity element of A'', and so $R(A) = A''$. $\square$